Mini Project: Analysis of the Iris Dataset in R

DTS 211 - Introduction to R Programming

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1 Introduction

The Iris dataset is one of the most famous datasets in statistics. It was collected by Edgar Anderson and made popular by Ronald Fisher in 1936. It comes built into R, so we do not need to download anything.

The dataset has 150 flowers. Each flower belongs to one of three species: setosa, versicolor and virginica. There are 50 flowers of each species. For every flower, four measurements were taken in centimetres: sepal length, sepal width, petal length and petal width.

The purpose of this project is to explore the data, draw some plots, work out simple summary statistics, and then build a model that can tell the species of a flower from its four measurements.

2 Methodology

The analysis was done in R in the following steps:

1. **Data import and exploration.** The data was loaded with `data(iris)`. Then `head()`, `str()` and `summary()` were used to see what the data looks like.
2. **Data visualisation.** Base R plots were used. A pair plot shows every pair of features at once, so it is a quick way to see which features separate the species. Boxplots compare each feature across the three species. Histograms show the shape of each variable. Finally, a scatter plot of petal length against petal width was drawn, because the pair plot suggested that the petal measurements separate the species best.
3. **Statistical summary.** The mean, median and standard deviation of each feature were calculated with `mean()`, `median()` and `sd()`.
4. **Classification.** The data was split into 70% training and 30% testing using `sample()` after `set.seed(123)`, so the results can be repeated. A k-nearest neighbours (k-NN) model with k = 5 was fitted using `knn()` from the `class` package. k-NN was chosen because it is simple to understand, needs no special assumptions about the data, and works well when the groups form clear clusters, which the plots showed. The model was checked with a confusion matrix and the accuracy on the test set.

3 Results and Discussion

3.1 Data import and exploration

```
# load the dataset
data(iris)

# first six rows
head(iris)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5          1.4         0.2  setosa
## 2          4.9         3.0          1.4         0.2  setosa
## 3          4.7         3.2          1.3         0.2  setosa
## 4          4.6         3.1          1.5         0.2  setosa
## 5          5.0         3.6          1.4         0.2  setosa
```

```
## 6          5.4         3.9          1.7         0.4  setosa
```

The first rows show the four measurement columns and the `Species` column. All the first rows are setosa because the data is sorted by species.

```
# structure of the data
str(iris)
```

```
## 'data.frame':    150 obs. of  5 variables:
##  $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
##  $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
##  $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
##  $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
##  $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
```

The data has 150 rows and 5 columns. The four measurements are numeric (`num`) and `Species` is a factor with 3 levels.

```
# summary of each column
summary(iris)
```

```
##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
##  Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
##  Median :5.800   Median :3.000   Median :4.350   Median :1.300
##  Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
##  Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
##        Species
##  setosa    :50
##  versicolor:50
##  virginica :50
##
##
##
```

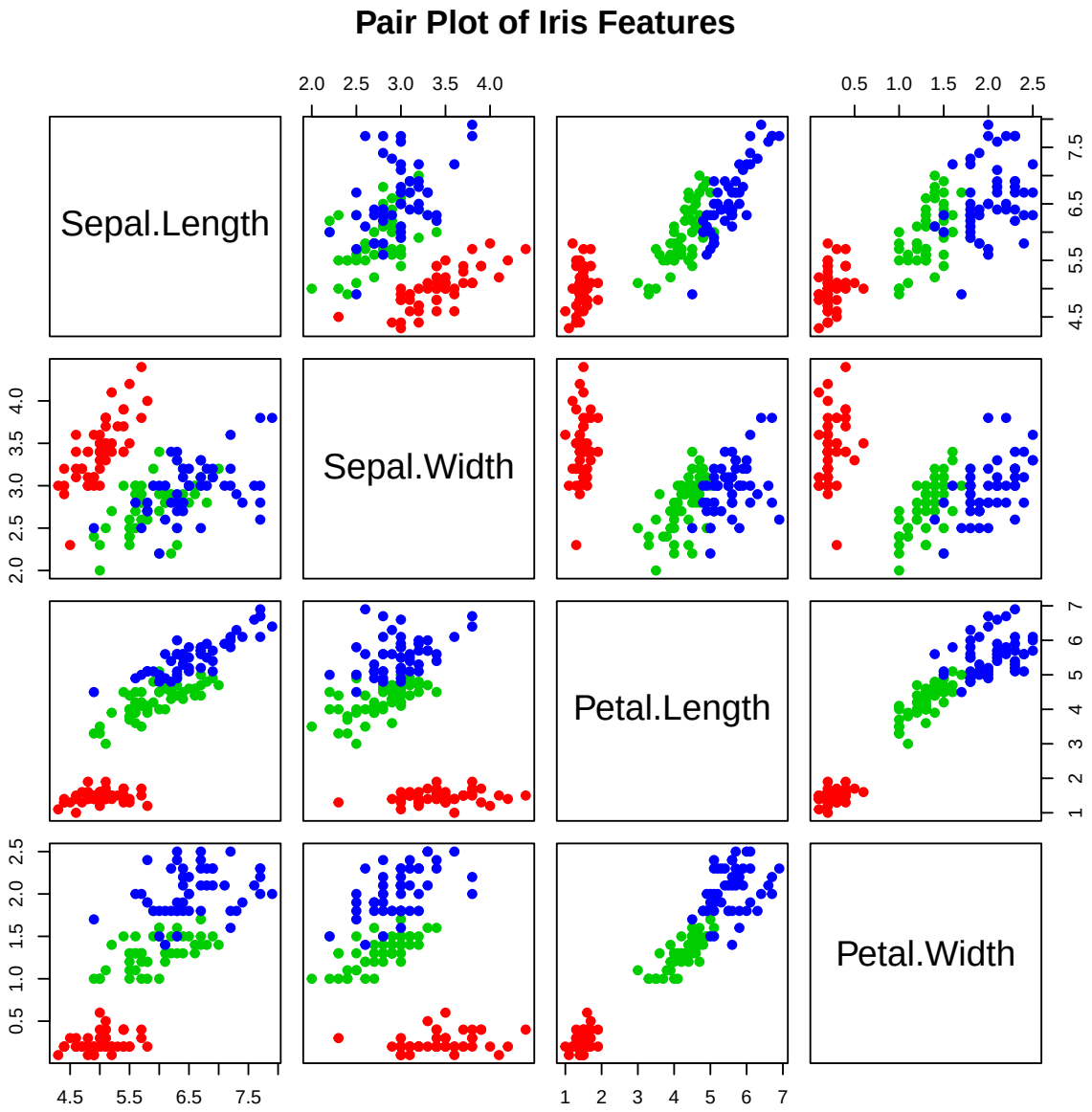
The summary shows there are no missing values and each species has exactly 50 flowers. Petal length has the biggest range, from 1.0 to 6.9 cm, while sepal width has the smallest range, from 2.0 to 4.4 cm.

3.2 Data visualisation

3.2.1 Pair plot

```
# colours for the three species
colours <- c("red", "green3", "blue")

# pair plot of the four features
pairs(iris[, 1:4], col = colours[iris$Species], pch = 19,
      main = "Pair Plot of Iris Features")
```



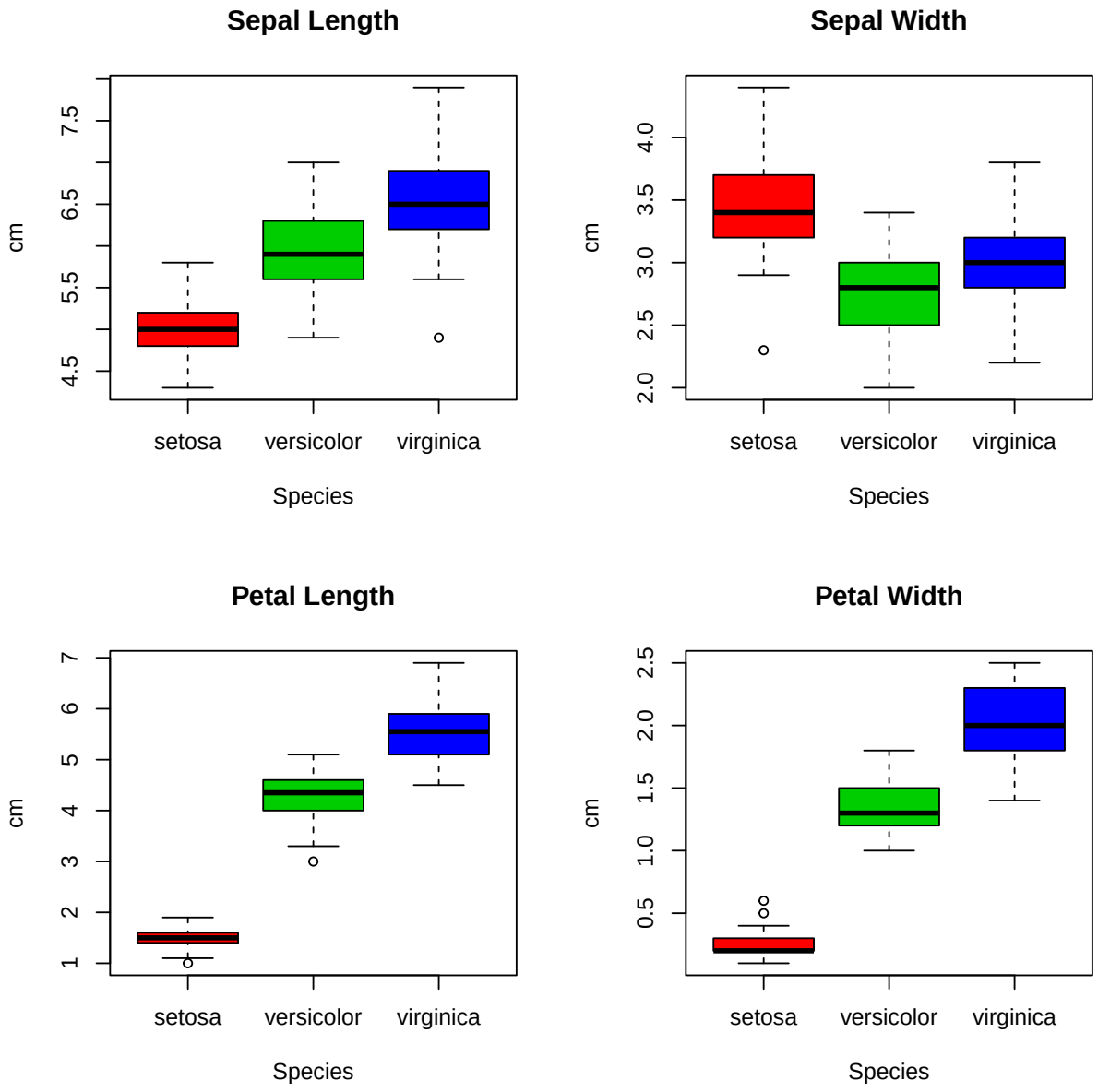
In the pair plot, red is setosa, green is versicolor and blue is virginica. Setosa (red) is clearly separated from the other two species in almost every panel. Versicolor and virginica overlap a bit, especially in the sepal plots, but they are much more separated in the petal plots.

3.2.2 Boxplots by species

```
# 2 by 2 grid of plots
par(mfrow = c(2, 2))

boxplot(Sepal.Length ~ Species, data = iris, col = colours,
        main = "Sepal Length", ylab = "cm")
```

```
boxplot(Sepal.Width ~ Species, data = iris, col = colours,
        main = "Sepal Width", ylab = "cm")
boxplot(Petal.Length ~ Species, data = iris, col = colours,
        main = "Petal Length", ylab = "cm")
boxplot(Petal.Width ~ Species, data = iris, col = colours,
        main = "Petal Width", ylab = "cm")
```




```
# back to one plot at a time
par(mfrow = c(1, 1))
```

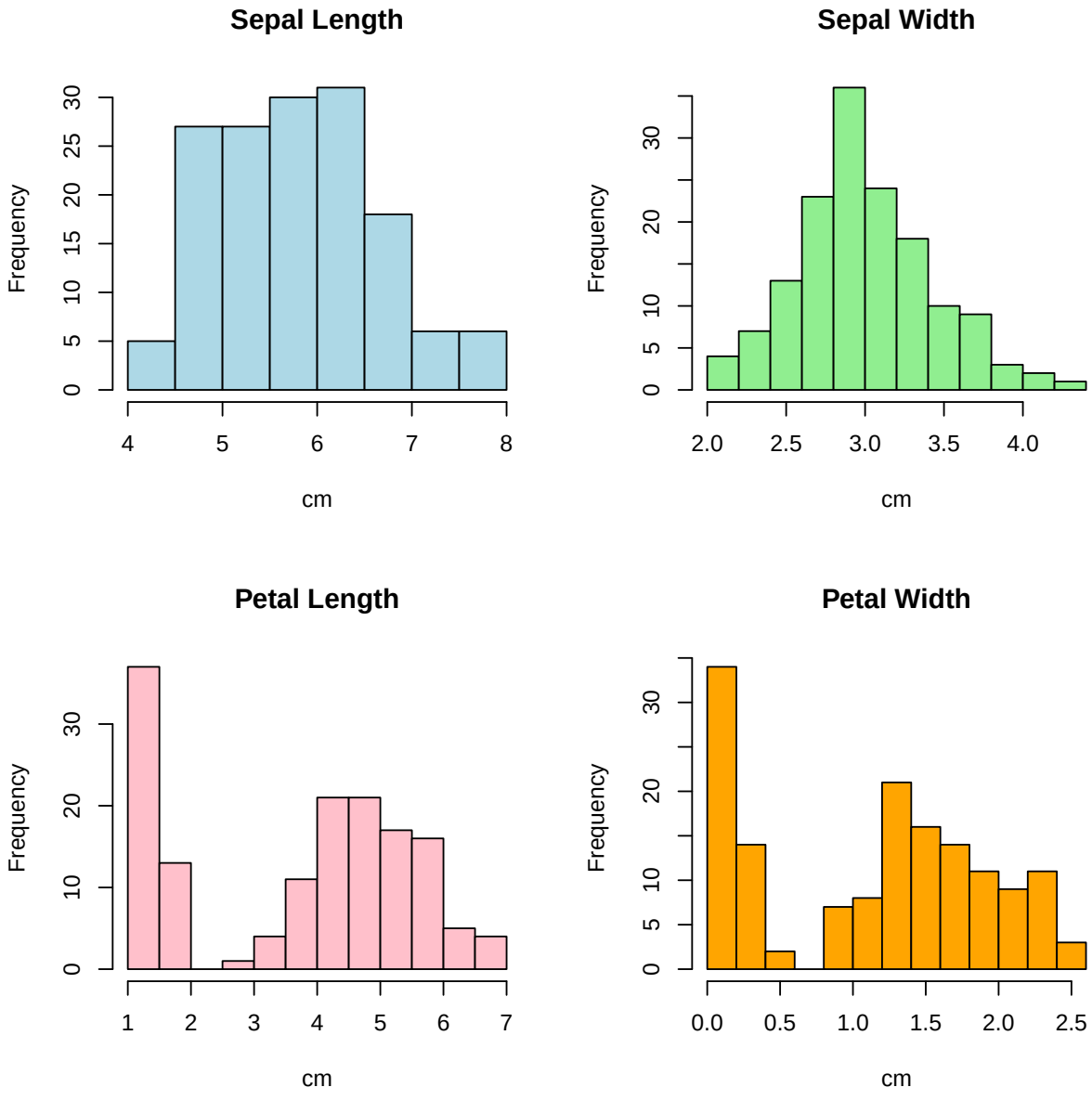
The boxplots show that virginica has the longest sepals and the biggest petals, and setosa has the smallest petals. Setosa is different for sepal width, because it has the widest sepals of the three. For

petal length and petal width, the boxes of the three species do not overlap at all, which means the petals are very useful for telling the species apart.

3.2.3 Histograms

```
par(mfrow = c(2, 2))

hist(iris$Sepal.Length, col = "lightblue",
     main = "Sepal Length", xlab = "cm")
hist(iris$Sepal.Width, col = "lightgreen",
     main = "Sepal Width", xlab = "cm")
hist(iris$Petal.Length, col = "pink",
     main = "Petal Length", xlab = "cm")
hist(iris$Petal.Width, col = "orange",
     main = "Petal Width", xlab = "cm")
```



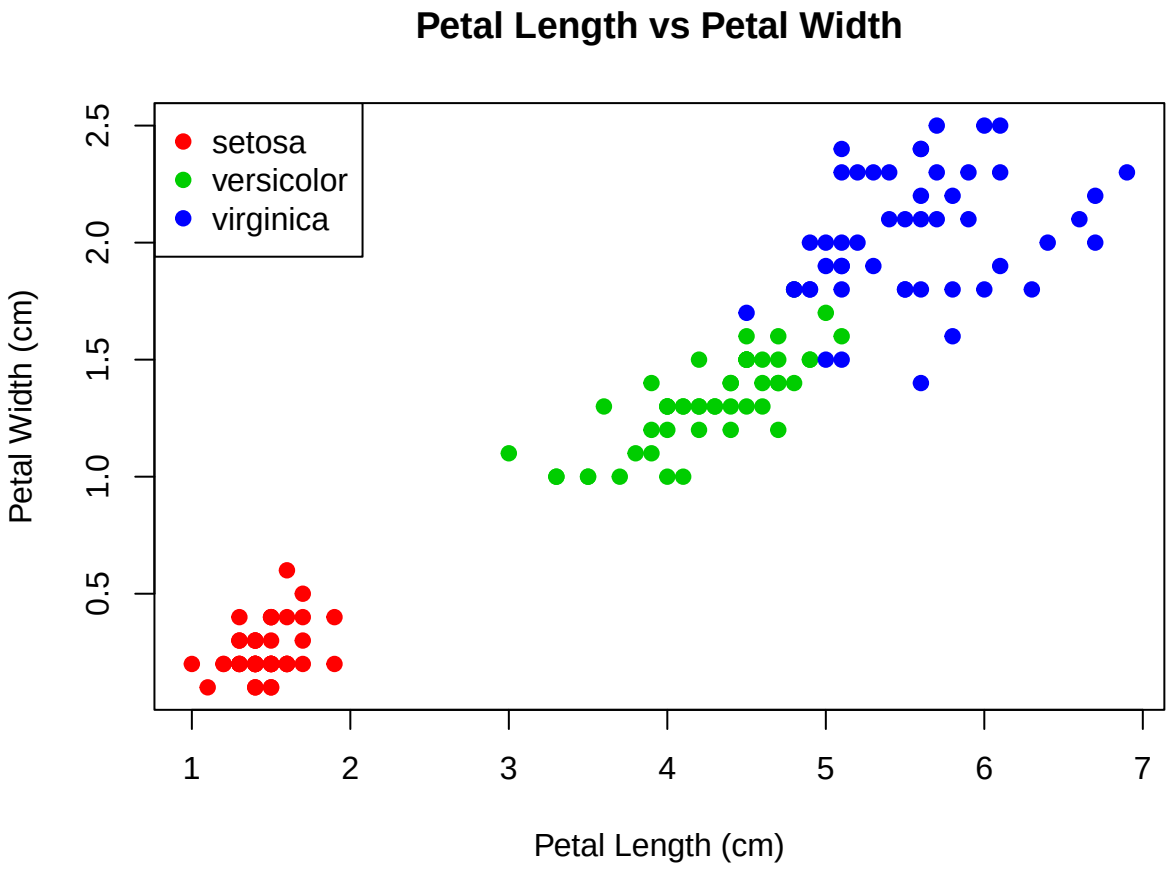

```
par(mfrow = c(1, 1))
```

Sepal length and sepal width look roughly bell shaped. Petal length and petal width are different: they have two separate groups. The small group on the left is the setosa flowers and the bigger group on the right is versicolor and virginica together.

3.2.4 Scatter plot of petal length and petal width

```
plot(iris$Petal.Length, iris$Petal.Width,
     col = colours[iris$Species], pch = 19,
     main = "Petal Length vs Petal Width",
     xlab = "Petal Length (cm)", ylab = "Petal Width (cm)")
```

```
legend("topleft", legend = levels(iris$Species),
       col = colours, pch = 19)
```



This plot shows the three species as three groups. Setosa sits alone in the bottom left corner. Versicolor is in the middle and virginica is at the top right, with only a small overlap between them. It also shows that flowers with longer petals also have wider petals.

3.3 Statistical summary

```
# mean of each feature
mean(iris$Sepal.Length)
```

```
## [1] 5.843333
```

```
mean(iris$Sepal.Width)
```

```
## [1] 3.057333
```

```
mean(iris$Petal.Length)
```

```
## [1] 3.758
```

```r
mean(iris$Petal.Width)
```

```
## [1] 1.199333
```

```r
# median of each feature
median(iris$Sepal.Length)
```

```
## [1] 5.8
```

```r
median(iris$Sepal.Width)
```

```
## [1] 3
```

```r
median(iris$Petal.Length)
```

```
## [1] 4.35
```

```r
median(iris$Petal.Width)
```

```
## [1] 1.3
```

```r
# standard deviation of each feature
sd(iris$Sepal.Length)
```

```
## [1] 0.8280661
```

```r
sd(iris$Sepal.Width)
```

```
## [1] 0.4358663
```

```r
sd(iris$Petal.Length)
```

```
## [1] 1.765298
```

```r
sd(iris$Petal.Width)
```

```
## [1] 0.7622377
```

To make it easier to read, the results are put in one table, rounded to 2 decimal places.

```r
means <- c(mean(iris$Sepal.Length), mean(iris$Sepal.Width),
           mean(iris$Petal.Length), mean(iris$Petal.Width))
medians <- c(median(iris$Sepal.Length), median(iris$Sepal.Width),
             median(iris$Petal.Length), median(iris$Petal.Width))
sds <- c(sd(iris$Sepal.Length), sd(iris$Sepal.Width),
         sd(iris$Petal.Length), sd(iris$Petal.Width))

stats <- data.frame(Mean = means, Median = medians, SD = sds)
rownames(stats) <- names(iris)[1:4]
round(stats, 2)
```

```
##              Mean Median   SD
## Sepal.Length 5.84   5.80 0.83
## Sepal.Width  3.06   3.00 0.44
## Petal.Length 3.76   4.35 1.77
```

```
## Petal.Width  1.20   1.30 0.76
```

We can also look at the mean petal measurements for each species.

```
# mean petal length and width for each species
tapply(iris$Petal.Length, iris$Species, mean)
```

```
##     setosa versicolor  virginica
##      1.462      4.260      5.552
```

```
tapply(iris$Petal.Width, iris$Species, mean)
```

```
##     setosa versicolor  virginica
##      0.246      1.326      2.026
```

Interpretation. Sepal length has the largest mean (5.84 cm) and petal width has the smallest (1.20 cm). For sepal length and sepal width, the mean and median are very close (5.84 and 5.80, 3.06 and 3.00), so these two variables are fairly symmetric.

Petal length is different. Its mean (3.76 cm) is much lower than its median (4.35 cm), and it has the largest standard deviation (1.77 cm). This happens because the setosa flowers have very short petals (mean 1.462 cm) and pull the mean down, while versicolor (4.260 cm) and virginica (5.552 cm) are much longer. Petal width shows the same pattern, with a mean of 1.20 cm and a median of 1.30 cm. Sepal width has the smallest standard deviation (0.44 cm), so it changes the least between flowers. The big spread in the petal measurements is a sign that they carry the most information about species.

3.4 Classification with k-NN

3.4.1 How k-NN works

k-nearest neighbours is a simple classification method. To classify a new flower, it measures the distance between that flower and every flower in the training data, using the four measurements. It then picks the k closest flowers (here $k = 5$) and looks at their species. The new flower is given the species that most of those 5 neighbours have. There is no real "training" step; the model just remembers the training data.

3.4.2 Splitting the data

```
library(class)

# set the seed so the split is the same every time
set.seed(123)

# pick 70% of the row numbers for training
train_rows <- sample(1:nrow(iris), 0.7 * nrow(iris))

train <- iris[train_rows, ]
test <- iris[-train_rows, ]

nrow(train)
```

```
## [1] 105
```

```
nrow(test)
```

```
## [1] 45
```

```
# how many of each species in train and test
table(train$Species)
```

```
##
##     setosa versicolor  virginica
##         36         32         37
```

```
table(test$Species)
```

```
##
##     setosa versicolor  virginica
##         14         18         13
```

The training set has 105 flowers and the test set has 45 flowers. The test set has 14 setosa, 18 versicolor and 13 virginica, so all three species are included.

3.4.3 Fitting the model

```
# features (first four columns) and labels
train_x <- train[, 1:4]
test_x <- test[, 1:4]
train_y <- train$Species
test_y <- test$Species

# k-NN with k = 5
pred <- knn(train = train_x, test = test_x, cl = train_y, k = 5)
```

3.4.4 Confusion matrix and accuracy

```
# confusion matrix
cm <- table(Predicted = pred, Actual = test_y)
cm
```

```
##             Actual
## Predicted    setosa versicolor virginica
##   setosa         14          0         0
##   versicolor      0         17         0
##   virginica       0          1        13
```

```
# accuracy = correct predictions / total predictions
accuracy <- sum(diag(cm)) / sum(cm)
accuracy
```

```
## [1] 0.9777778
```

```r
# as a percentage
round(accuracy * 100, 2)
```

```
## [1] 97.78
```

Interpretation. In the confusion matrix, the rows are the predicted species and the columns are the actual species. The numbers on the diagonal are the correct predictions. The model got all 14 setosa and all 13 virginica flowers right. Out of the 18 versicolor flowers, 17 were correct and 1 was wrongly predicted as virginica.

So the model got 44 out of 45 test flowers right, which is an accuracy of 97.78%. This is a very good result. The only mistake was between versicolor and virginica, which matches what we saw in the plots: these two species overlap a little, while setosa is completely separate.

4 Conclusion

This project explored the Iris dataset using base R. The data is clean, with 150 flowers, 50 per species and no missing values.

The plots and statistics showed that petal length and petal width are the most useful features. Setosa is easy to tell apart because its petals are much smaller. Versicolor and virginica are similar and overlap slightly.

The k-NN model with k = 5 reached an accuracy of 97.78% on the test set, making only 1 mistake out of 45 flowers. This shows that a simple method can classify iris species very well from the four measurements.

Recommendations for future analysis:

- Try different values of k (for example 1, 3, 7 and 9) and compare the accuracy.
- Scale the features before using k-NN, so all variables count equally in the distance.
- Use cross-validation instead of one split, because a test set of 45 flowers is small and the accuracy could change with a different seed.
- Compare k-NN with other methods such as decision trees or logistic regression.
- Test whether using only the two petal features gives the same accuracy.